SIA: Smartphone Interfaces Assistant using ChatGPT

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- Inspiration & Problem Statement
- Research Questions
- Solution Overview - SIA

Frontend Design

Key Features:

- Easy Access: A floating button located at the bottom-right corner.
- Two Options: Text input or automatic screenshot and upload.
- Simplified Navigation: Eliminates the need for complex menus.



Frontend Design

Once tapped, the floating button provides two options:

- Text Input:
  - Opens a small chat window where users can type questions.
  - Users can scroll through the chat history and close the window using the “X” button.
- Automatic Screenshot Upload:
  - Automatically captures and uploads the current screen.
  - Users can scroll through the chat history and close the window using the “X” button.



Frontend Design

- Small Chat Window:
  - Includes an "Expand" icon in the top-left corner to switch to fullscreen.
  - Ideal for quick interactions.
- Full Screen Mode:
  - Allows users to scroll through chat history and navigate content comfortably.
  - A "Return" button in the top-left corner lets users switch back to the smaller window.

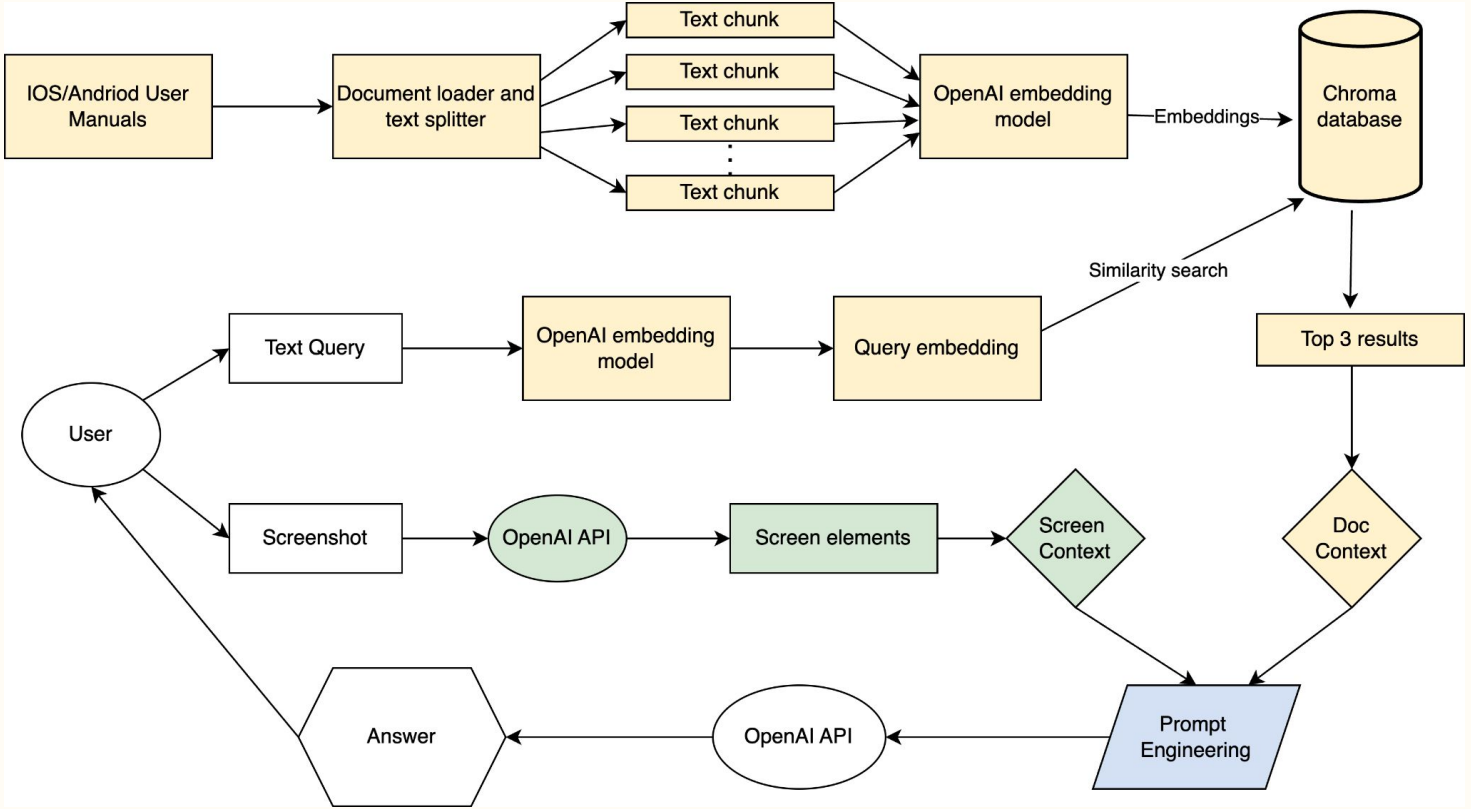


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Backend Design



Backend Design

```
PROMPT_TEMPLATE = """
Hi, I am an older adult who need your assistance
to understand how to do some tasks on my phone.
Based on the following context and current screen information:

Screen Context:
{screen_context}

Knowledge Base Context:
{context}

---

Please help answer this question: {question}

Provide step-by-step instructions from my current screen if applicable,
referencing specific UI elements visible on the current screen.
Please provide easy-to-follow answers and avoid any technical jargons.
"""
```

User Study

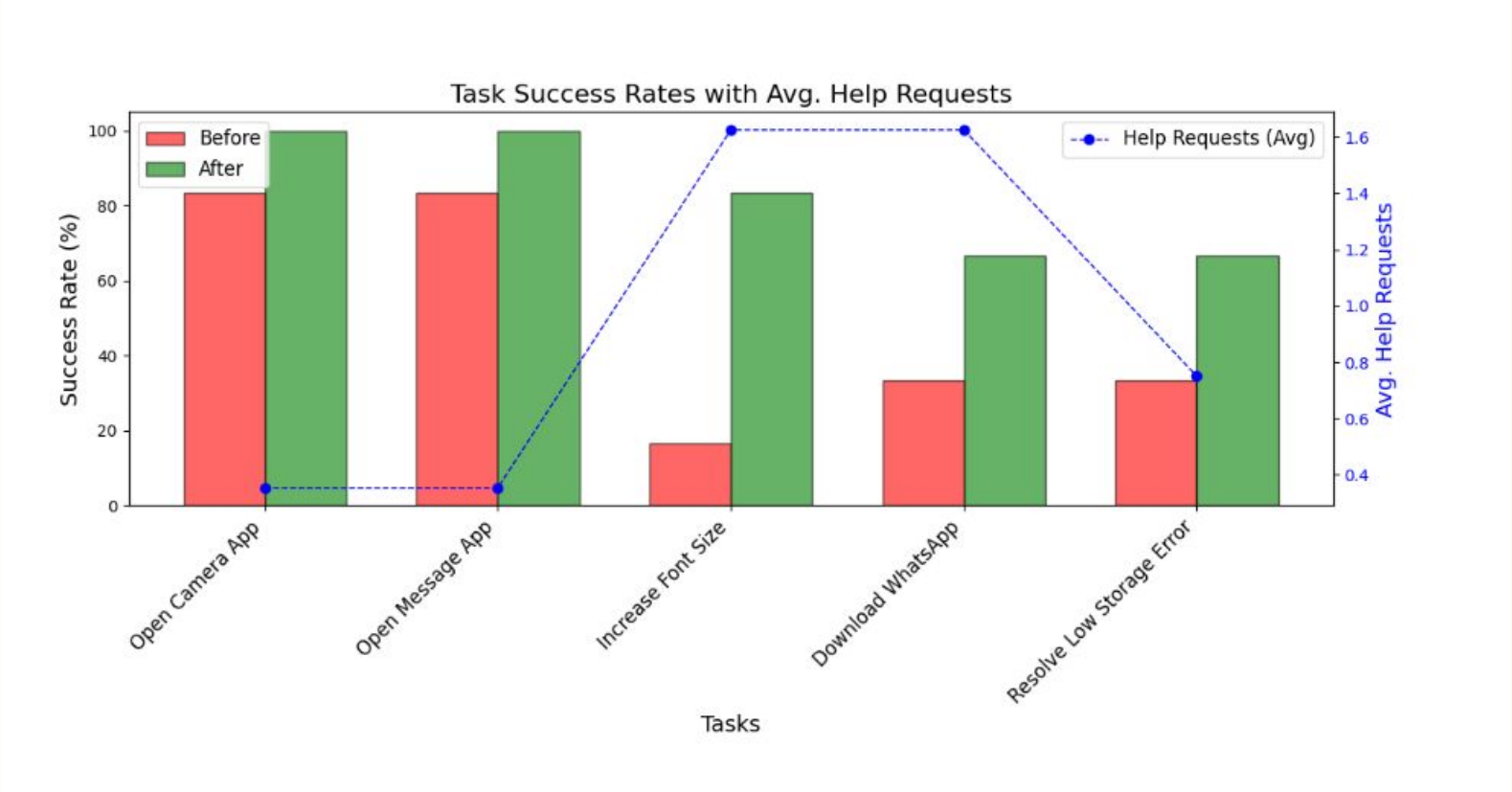
- Validate the Design Concept
- Test Effectiveness
- Gather User Feedback

Tasks & Evaluation metrics

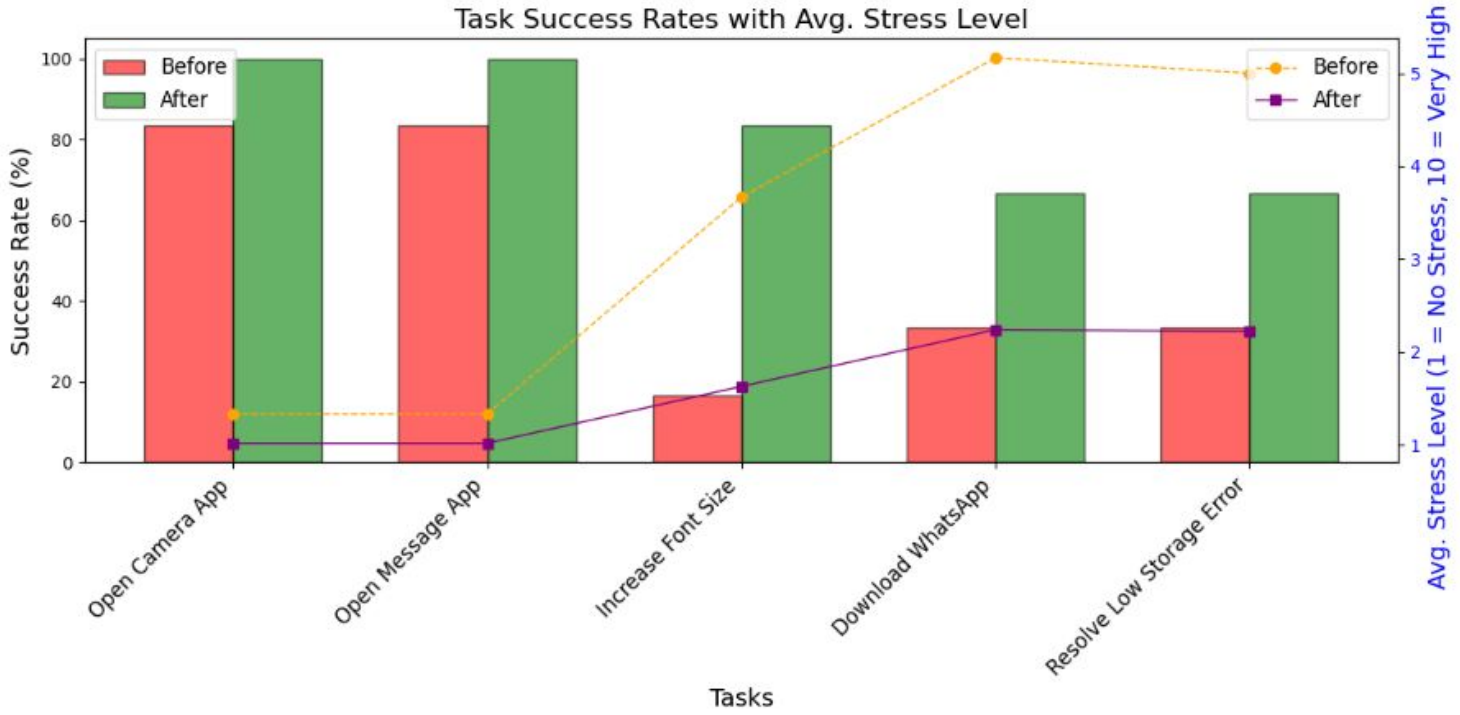
- Open the Camera app
- Open the Messages app
- Increase font size
- Download app (ex. WhatsApp)
- Manage phone storage

- Task difficulty
- Success rate
- Help requests
- Mental demand
- Stress level

Results


Task Success Rates with Avg. Help Requests
Before
After
Help Requests (Avg)
Success Rate (%)
Avg. Help Requests
Tasks
Open Camera App
Open Message App
Increase Font Size
Download WhatsApp
Resolve Low Storage Error
0
20
40
60
80
100
0.4
0.6
0.8
1.0
1.2
1.4
1.6

Results


Task Success Rates with Avg. Stress Level
Success Rate (%)
Avg. Stress Level (1 = No Stress, 10 = Very High
Tasks
Before
After
Before
After
0
20
40
60
80
100
1
2
3
4
5
Open Camera App
Open Message App
Increase Font Size
Download WhatsApp
Resolve Low Storage Error

Conclusion & Future Work

Forecast Depression Level and Risk of Suicide

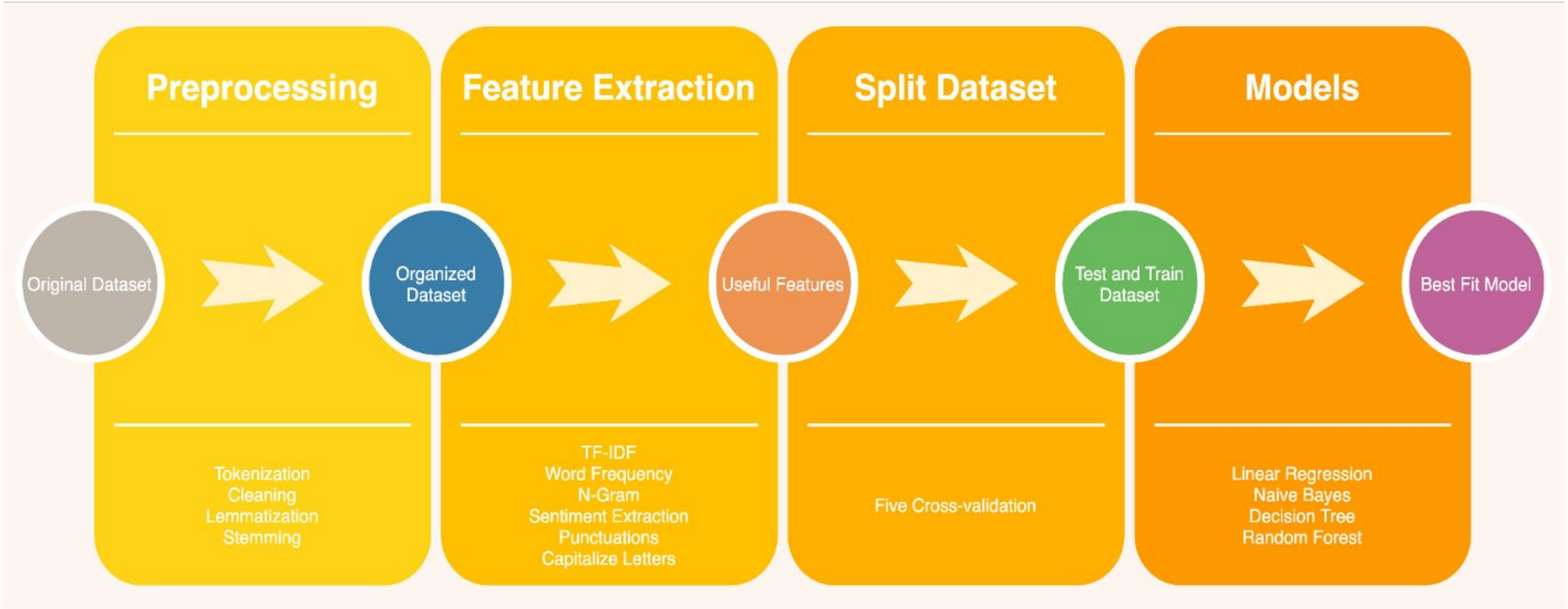
Ziyu Xu

Dataset

	User	Post	Label
0	user-0	['Its not a viable option, and youll be leavin...	Supportive
1	user-1	['It can be hard to appreciate the notion that...	Ideation
2	user-2	['Hi, so last night i was sitting on the ledge...	Behavior
3	user-3	['I tried to kill my self once and failed badl...	Attempt
4	user-4	['Hi NEM3030. What sorts of things do you enjo...	Ideation
...	...	...	...
495	user-495	['Its not the end, it just feels that way. Or ...	Supportive
496	user-496	['It was a skype call, but she ended it and Ve...	Indicator
497	user-497	['That sounds really weird.Maybe you were Dist...	Supportive
498	user-498	['Dont know there as dumb as it sounds I feel ...	Attempt
499	user-499	['>It gets better, trust me.Ive spent long ...	Behavior

500 rows × 3 columns

Methods


Preprocessing
Feature Extraction
Split Dataset
Models
Original Dataset
Organized Dataset
Useful Features
Test and Train Dataset
Best Fit Model
Tokenization
Cleaning
Lemmatization
Stemming
TF-IDF
Word Frequency
N-Gram
Sentiment Extraction
Punctuations
Capitalize Letters
Five Cross-validation
Linear Regression
Naive Bayes
Decision Tree
Random Forest

Thank you!!